Limitations of the ECQC Inequality: A Counterexample in Composite Dimension $d = 4$

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Introduction

- **Is Quantum correlation stronger than Classical correlation?**
- **CQC conjecture tries to answer this question.**
- **ECQC extends CQC to multiple measurements and higher dimensions.**
- **Relates classical measurement correlations to total quantum mutual information.**

Motivation and Problem

- The CQC inequality [1] asks: Is the sum of two classical correlations at most the total quantum correlation?
- ECQC conjecture [2]:

$$I(A:B) \geq \min_{S\in[d+1]_d} \sum_{i\in S} I(M_i^A : M_i^B)$$

- Proven for isotropic states and prime dimensions.
- Unknown for composite dimensions.

$\Rightarrow$ **Does ECQC hold in $d = 4$?**

Preliminaries

- In mutually unbiased bases (MUBs), measurement in one basis provides no information about other measurement bases.
- Maximally entangled states yield maximal correlations between two parties.

Measurements and State

- We use a complete set of 5 mutually unbiased bases from [3] in $d = 4$.

Maximally entangled state in $d = 4$:

$$|\Phi_4\rangle = \frac{1}{2}(|00\rangle + |11\rangle + |22\rangle + |33\rangle)$$

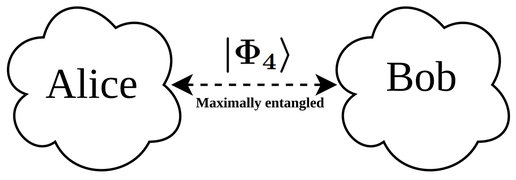


Figure 1:Alice and Bob share maximally entangled state

Proof Sketch

- We consider the state $|\Phi_4\rangle$

Reduced states: $\rho_A = \rho_B = \frac{I_4}{4}$ (maximally mixed)

Entropies: $S(A) = S(B) = 2, \quad S(AB) = 0$

Mutual information: $I(A : B) = S(A) + S(B) - S(AB) = 2 + 2 - 0 = 4$

- Measure both systems (A, B) in the same basis U_i, for all $i \in \{1, 2, 3, 4, 5\}$
- Apply Born rule for probabilities to $|\Phi_4\rangle$:

$$P_U(a,b) = |\langle u_a| \otimes \langle u_b|\Phi_4\rangle|^2 = \boxed{\tfrac{1}{4}|(U^TU)_{ab}|^2}$$

Important Result

The prime-dimensional ECQC Conjecture does not hold for composite d=4.

Proof Sketch (cont.)

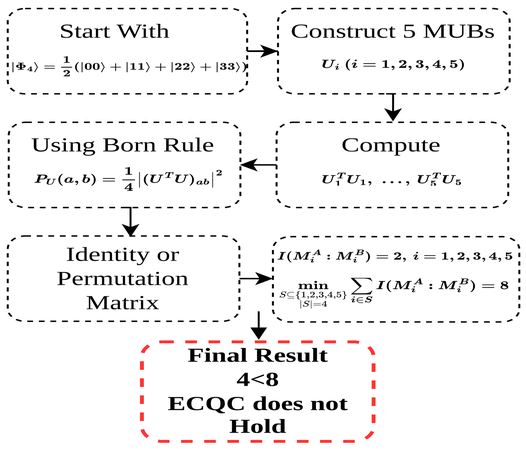


Figure 2:A schematic view of our proof method

Key Calculations

$$U_1^TU_1 = U_2^TU_2 = I_4 = \textbf{Identity matrix}$$

$$U_3^TU_3 = U_4^TU_4 = U_5^TU_5 = \begin{pmatrix} 0&1&0&0 \\ 1&0&0&0 \\ 0&0&0&1 \\ 0&0&1&0 \end{pmatrix}$$

$$U_3^TU_3 = U_4^TU_4 = U_5^TU_5 = \textbf{Permutation matrix}$$

- The matrices U_i correspond to mutually unbiased bases.

Measurement Outcome Structure

Measurement outcomes from $U_i^TU_i$:

Identity case: $(0,0),\ (1,1),\ (2,2),\ (3,3)$

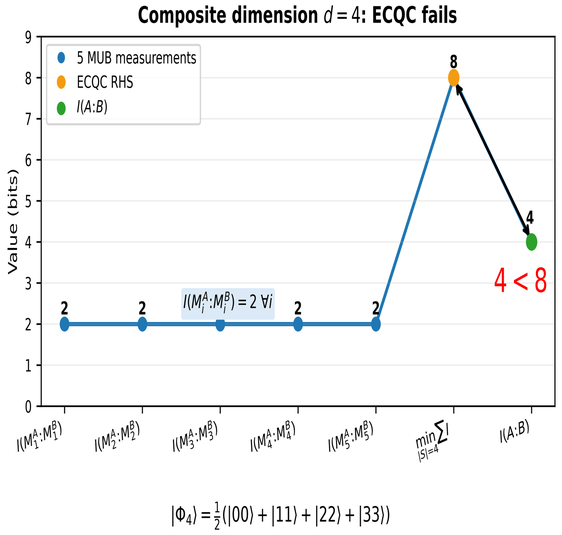
Permutation case: $(0,1),\ (1,0),\ (2,3),\ (3,2)$

$$P(a,b) = \frac{1}{4}$$

Interpretation:

- $P(a,b)$ denotes the joint probability that Alice obtains outcome a and Bob obtains outcome b.
- For example, $(0,0)$ means both obtain outcome 0.
- Only 4 outcome pairs occur, each with equal nonzero probability.

ECQC violation in $d = 4$



$|\Phi_4\rangle = \frac{1}{2}(|00\rangle + |11\rangle + |22\rangle + |33\rangle)$

Figure 3:ECQC violation in composite dimension $d = 4$

Mutual Information Values

Entropies: We compute $H(M_i^A) = H(M_i^B) = 2, \quad H(M_i^A, M_i^B) = 2$

Mutual information: $I(M_i^A : M_i^B) = H(M_i^A) + H(M_i^B) - H(M_i^A, M_i^B) = 2 + 2 - 2 = 2$, for all $i \in \{1, 2, 3, 4, 5\}$

- **All measurements give maximal classical correlation.**

Main Result

Right-hand side of the ECQC:

$$\min_{\substack{S\subseteq\{1,\ldots,5\} \\ |S|=4}} \sum_{i\in S} I(M_i^A : M_i^B) = 8$$

Left-hand side of the ECQC:

$$I(A:B) = 4$$

$$4 < 8$$

ECQC inequality fails in $d = 4$

Conclusion

- The ECQC extension does not generalize to composite dimensions.
- All MUB measurements contribute equally in $d = 4$, unlike in $d = 3$.
- The dimensional structure behaves fundamentally differently in prime and composite dimensions.

Future Work

- Explain the difference between $d = 3$ and $d = 4$: why all five MUBs yield non-zero mutual information in $d = 4$, while only two do so in $d = 3$.
- Develop a generalized ECQC inequality valid for all bipartite quantum states.
- Prove the CQC conjecture for general bipartite systems.

References

[1] James Schneeloch, Curtis J Broadbent, and John C Howell.
Uncertainty relation for mutual information.
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[2] Hasan Iqbal.
On the cqc conjecture.
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[3] Stephen Brierley.
Mutually unbiased bases in low dimensions.
PhD thesis, University of York, 2009.